

OSHADEE KAUSHALYA

Graduate Software Developer — Software Engineering Researcher

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Professional Summary

Master of Software Engineering graduate with a background in software development, artificial intelligence, and technology education. Experienced in building web and mobile applications using JavaScript, TypeScript, React, Python, Flask, SQL, and modern development tools. Passionate about creating practical software solutions, rapidly prototyping ideas, and leveraging AI-assisted development tools such as Claude Code, GitHub Copilot, and Amazon Q. Strong communicator with experience translating technical concepts into clear solutions for diverse users through 4+ years of tertiary teaching and project supervision.

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, SQL, PHP, C

Web Technologies and Frameworks: React.js, Flask, HTML5, CSS3, Tailwind CSS, Bootstrap

Mobile Development: Flutter, Android Studio

AI & Automation Tools: Claude- Code, GitHub Copilot, OpenAI API

Cloud & Databases: AWS, Microsoft Azure, MySQL, SQLite

Tools & Practices: Figma, Docker, Git, GitHub, VS Code

Relevant Projects

ResQNet — Flutter, Python, TensorFlow Lite

- Designed and developed an AI-powered disaster communication platform capable of autonomously detecting distress situations using on-device machine learning and delivering emergency alerts when conventional communication infrastructure is unavailable.
- Trained and optimized a Convolutional Neural Network (CNN) using TensorFlow and TensorFlow Lite to classify distress-related audio events from real-world environmental conditions, achieving approximately 88% detection accuracy.
- Engineered a Flutter mobile application integrating AI inference, location services, emergency contact management, and offline communication mechanisms to improve emergency response capabilities.

Rescue Coordination Dashboard — React.js, Tailwind CSS, Firebase, JavaScript

- Built a real-time web dashboard for emergency responders by integrating React and Firebase for live data sync, delivering interactive map-based alert tracking and priority-ranked victim monitoring.
- Translated UI/UX requirements into clean, reusable React components styled with Tailwind CSS, matching the fast-iteration workflow of a fast-moving research project.

TravelMate — Flask, JavaScript, Bootstrap, OpenAI API, Docker, Azure

- Co-developed an intelligent travel planning platform that generates personalized travel recommendations using large language models and user preference data.
- Integrated OpenAI APIs to automate itinerary generation, destination recommendations, and contextual travel assistance through natural language interactions.
- Containerized the application using Docker and deployed cloud-hosted environments on Microsoft Azure to support scalable development and testing workflows.
- Developed AI-assisted workflows using OpenAI APIs and prompt engineering techniques to automate personalised itinerary generation and user interactions.
- Utilized AI-powered development tools, automated testing, PyTest, and code-quality practices to improve development efficiency and maintainability.

Professional Experience

Assistant Lecturer/ Instructor — University of Colombo School of Computing

Nov 2022 – Jul 2025

Positions: - [Profile](#)

1. Temporary Assistant Lecturer – November 2022 – July 2025

2. Instructor - February 2022 – October 2022

- Delivered high-quality undergraduate instruction in Data Structures & Algorithms and Software Engineering to 200+ students per term using practical Python and Java demonstrations.
- Taught Advanced Data Structures, Algorithms, and Software Engineering principles to 200+ students per term, demonstrating production-ready Java, Python, and Object-Oriented Design practices.

- Mentored and evaluated 10+ student engineering teams through full software development life cycles(SDLC), conducting architecture reviews and enforcing Git workflow and code quality standards.

Assistant Lecturer / Tutor — ESOF T Metro Campus

Apr 2021 – Feb 2022

Positions:

1. Assistant Lecturer – December 2021 – February 2022
 2. Tutor- April 2021 – November 2021
- Delivered practical programming lectures and lab sessions for ICT diploma-level students, focusing on core C programming, Object-Oriented Programming (OOP), and application development.
 - Designed and built a comprehensive hands-on curriculum for a Mobile Application Development certificate course covering Java, Flutter, and REST API integration.
 - Served as Lecturer-in-Charge for Higher National Diploma (HND) software modules, overseeing project milestones, technical evaluations, and student progress.

Software Engineering Intern — ACECAM (Pvt) Ltd

Oct 2020 – Apr 2021

- Engineered and maintained enterprise software modules for a live, commercial ERP system using corporate software engineering practices.
- Developed and tested web-based application features using PHP, JavaScript, HTML, CSS, Bootstrap, and SQL technologies to support business operations and asset management processes.
- Worked within a professional engineering team, gaining hands-on experience with real-world software delivery practices.
- Gained practical experience with version control, software testing, database management, and professional software delivery practices within a commercial technology environment.

Education

Master of Software Engineering - NZQA Level 09

Yoobee College of Creative Innovation, New Zealand · 2025 – 2026

Focusing on Artificial Intelligence, Disaster Communication Systems, Mobile Computing, and Software Engineering research.

Bachelor of Information & Communication Technology (Honours) - NZQA Level 08

University of Kelaniya, Sri Lanka · 2017 – 2021

First Class Honours, GPA 3.71/4.00 · Software Systems Technology

Referees

Referees available upon request